Python Engineering - Class 07

NumPy Arrays | EX15 + EX15b - NumPy Arrays

Class objective: Move from Python collections to multidimensional numerical structures and learn shape, ndim, axes and vectorized operations.

This document is the base exercise guide. It is not a full solution to copy: it organizes the problem, implementation steps, reference values and required validation.

Path

```
01_python_bootcamp/
`-- week_02_data_structures/class_07_numpy_arrays/
    |-- notes/
    `-- exercices/
```

Required workflow

```
PROBLEM
INPUTS
OUTPUTS
EQUATIONS
SPECIAL CASES
DO I NEED ITERATION?
ALGORITHM
VALIDATION
```

Before coding, write these eight sections. The goal is to separate problem reasoning from syntax.

EX15 - Trajectory Analysis with NumPy

Problem: Represent a trajectory as a numerical matrix and compute quantities across complete columns.

Reference inputs: trajectory rows [h,v,t]

Equations / relationship: specific energy = g*h + 0.5*v**2

Expected check: Matrix shape is (N,3); energy vector has N elements; compare selected rows with original states.

Implementation guide

- 1. Convert data to np.array.
- 2. Inspect shape/ndim/dtype.
- 3. Select columns.
- 4. Apply vectorized operations.
- 5. Analyze energy.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

EX15b - Shapes and Operations

Problem: Practice reshape, transpose, broadcasting, stacking and reductions along axes.

Reference inputs: small reference arrays

Equations / relationship: matrix/vector operations

Expected check: Understand (3,) vs (1,3), axis=0 vs axis=1, * vs @.

Implementation guide

- 1. Create 1D/2D arrays.

- 2. Inspect shape and ndim.
- 3. Reshape.
- 4. Stack arrays.
- 5. Compare element-wise and matrix multiplication.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

Final validation

Answer these questions as comments at the end of the exercise. The goal is not memorized wording but demonstrating that you can explain the code flow.

- Difference between shape, ndim and size.
- Why can a physical 3D vector have ndim=1?
- Difference between A[0] and A[:,0].
- Difference between * and @.
- What does axis=0 mean for an (N,3) matrix?
- What is broadcasting?

Completion criterion

The class is complete when the code runs, the reference checks make sense, you can explain every block without reading a solution, and you can modify at least one input while predicting the expected effect.